

STAT!

3D Model Overview

Step-by-step Treatment Aid Tool

Hardware & Architecture

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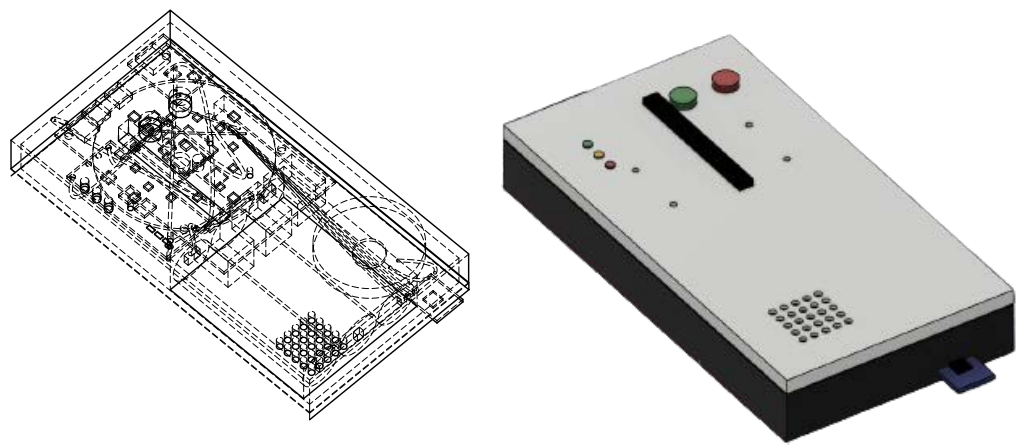
STAT!

Step-by-step Triage Aid Tool

3D Model Overview & Component Specification

(Note that all 3D Models shown are made by the team, no open source models were used.)

Complete Device Assembly — Isometric View



Document: STAT-3D-01 | **Date:** 07 May 2026 | **Version:** 5.1

Project: STAT! Emergency Medical AI | **Status:** Final

Executive Summary

The STAT! device is a compact, offline-capable emergency medical AI appliance housed in a 161 × 80 × 25.5 mm ABS plastic enclosure, designed to fit within DHA-compliant first-aid kits across the United Arab Emirates. The 3D model (Stat_Electronic.obj) represents the complete assembly with all subsystems: enclosure, CPU board, audio components, display module, battery pack, and interconnects. This document provides comprehensive specifications, component inventory, and integration details derived from the CAD model and technical drawings.

3D Model Statistics

Metric	Value	Details
Vertices	45,374	Geometric points defining model surfaces
Faces	90,524	Triangular or polygonal surface elements
Normal Vectors	50,608	Surface orientation data for rendering
Texture Coordinates	52,366	UV mapping for material application
Unique Groups	94	Distinct assemblies and sub-components
File Format	WaveFront OBJ	Generated by Autodesk ATF
Total File Size	~9.3 MB	High-resolution production model

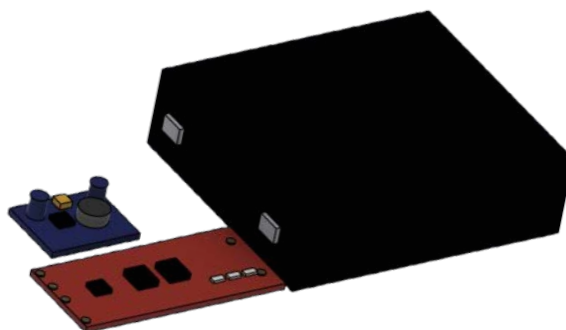
Component Breakdown & 3D Visualization

The STAT! assembly comprises ten primary subsystems, each represented as a distinct group in the 3D model. The subsystems are organized into three vertical zones to maximize internal volume utilization within the space-constrained envelope.

Zone A — Power & Battery Base (Height: 13–15 mm)

Exploded View — Battery & Power Components

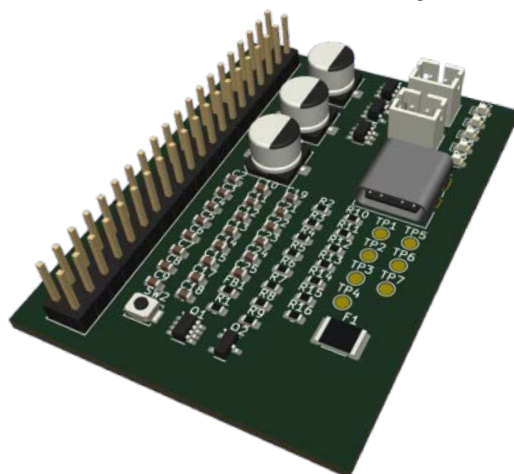
Battery Pack, BMS, Buck Converter Assembly



Component	Part No.	Material	Dimensions	Function
LiPo Battery (2S)	STAT-2026-006	Li-Polymer	57×68×13mm	5000mAh, 37Wh power source
Battery Mgmt System	STAT-2026-BMS	FR4 PCB	50×22mm	Charging, protection
Buck Converter	STAT-2026-BCK	FR4 PCB	22×17mm	7.4V→5V/3A regulated

Zone B — Compute & Processing (Height: 5–7 mm)

CPU Board Assembly

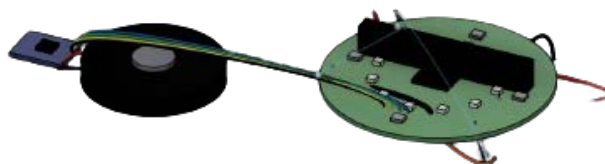


Component	Part No.	Material	Dimensions	Function
CPU Board (RPI5)	STAT-2026-003	FR4 PCB	86×56mm	Mistral 7B inference, system control

Zone C — Audio, Display & Interface Components

Exploded Assembly View

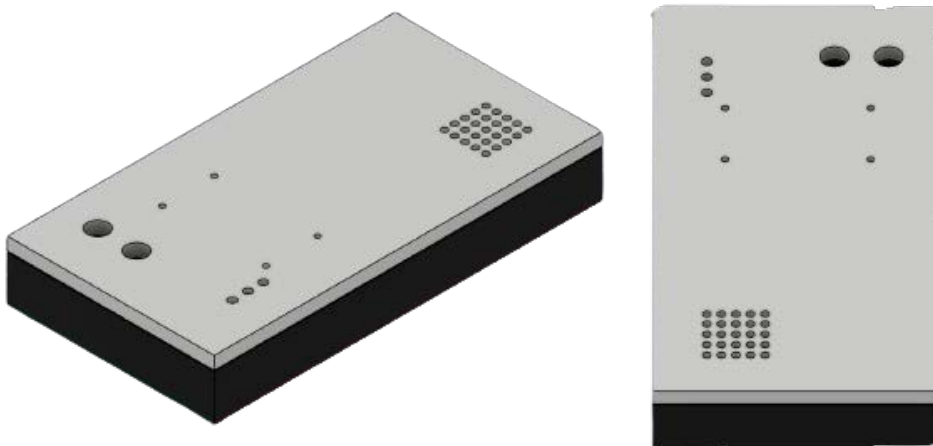
Speaker, Microphone, OLED, Flex PCB Assembly



Component	Part No.	Material	Dimensions	Function
Speaker Driver	STAT-2026-004	ABS/Neo	Ø40×13mm	Audio output, 3W Class D amp
Microphone	STAT-2026-005	ABS	Ø65 cutout	Voice input, INMP441 I2S
OLED Display	STAT-2026-007	FR4 PCB	26×25×18mm	Status display, SSD1306
Flex PCB	STAT-2026-010	Kapton	Custom	Signal & power routes

Enclosure & Hardware

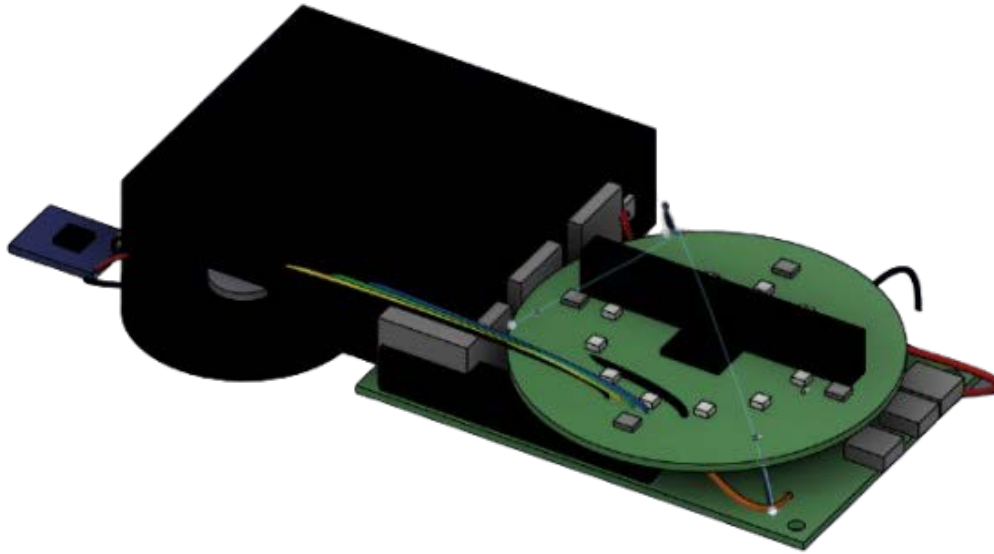
Enclosure Detail View — External Features & Port Locations



Component	Part No.	Material	Dimensions	Function
Enclosure Body	STAT-2026-001	ABS	161×80×25.5mm	Primary housing
Bottom Cover	STAT-2026-002	ABS	150×80mm	Base seal
Speaker Grille	STAT-2026-009	ABS	Ø65 aperture	Acoustic cover
USB-C Bezel	STAT-2026-008	ABS	~20mm zone	Charging port

System Integration & Internal Architecture

Internal Cross-Section View — Vertical Stack Layout



Subsystem Details & Signal Routing

Battery Pack & Power Distribution

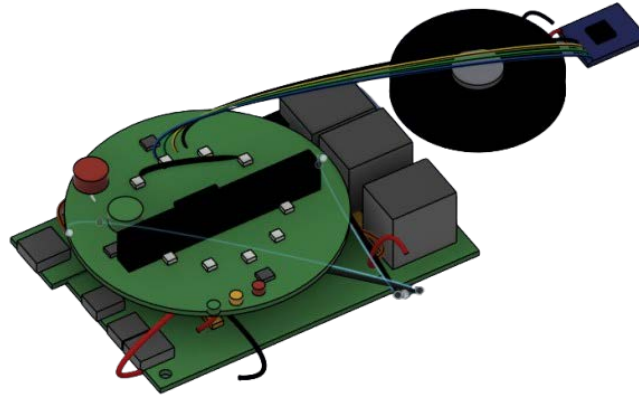
Battery Assembly Detail — Internal PCB Layout



Configuration: 2S1P (two 3.7V cells in series) | **Voltage:** 7.4V nominal (8.4V max, 6.0V cutoff)
Capacity: 5,000 mAh | **Energy:** 37.0 Wh | **Usable:** 31.45 Wh (85% DoD)
Discharge: 1C = 5A min | **Peak Load:** 1.51A | **Dimensions:** 57×68×13 mm

Audio Subsystem Architecture

Audio Signal Flow Diagram & Component Placement



Microphone: INMP441 MEMS omnidirectional, I2S interface, SNR 65 dB

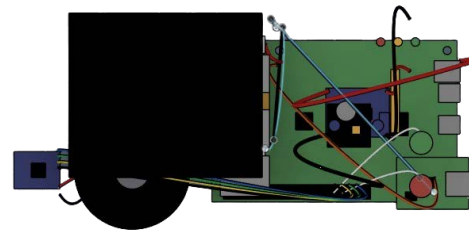
Speaker: Ø40 mm neodymium driver, 3W Class D amplifier (PAM8403)

Grille: Ø65 mm aperture with 12.5 mm annular gap for acoustic diffusion

CPR Metronome: 100 BPM procedural beat, 50 ms tone @ 800 Hz

Flex PCB & Signal Routing

Flex PCB & Trace Routing Diagram

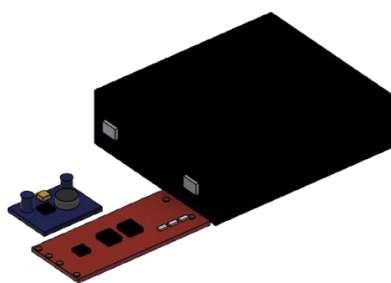


Power Routes: Battery (+7.4V) → BMS → Buck → 5V rail (CPU/amp) & 3.3V rail (OLED/mic)

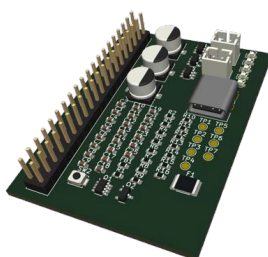
Signal Routes: I2S audio (shielded) | I2S mic in (shielded) | I2C OLED (400 kHz)

Complete Assembly Overview & Reference

Complete Device Disassembly



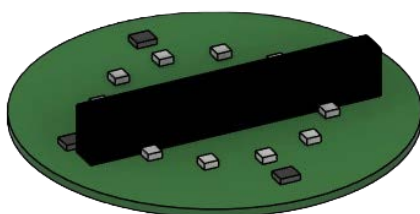
LiPo Battery, BMS, & Buck Converter



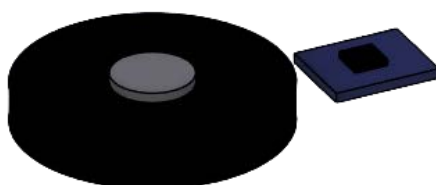
CPU Board



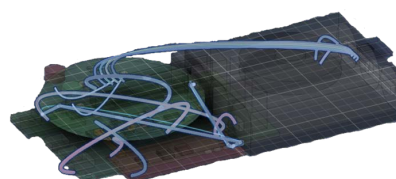
Enclosure



Microphone Module



Speaker



Wiring

Hardware Bill of Materials (Complete)

Item	Part No.	Qty	Description	Material
1	STAT-2026-001	1	Main Enclosure Body	ABS Plastic
2	STAT-2026-002	1	Bottom Cover Panel	ABS Plastic
3	STAT-2026-003	1	CPU Board	FR4 PCB (RPI5)
4	STAT-2026-004	1	Speaker Driver Unit	ABS/Neodymium
5	STAT-2026-005	1	Microphone Module	ABS Plastic
6	STAT-2026-006	1	LiPo Battery Pack	Li-Polymer
7	STAT-2026-007	1	OLED Display	FR4 PCB
8	STAT-2026-008	1	USB-C Port Bezel	ABS Plastic
9	STAT-2026-009	1	Speaker Grille Cover	ABS Plastic
10	STAT-2026-010	1	Flex PCB & Cable	Polyimide
11	STAT-2026-BMS	1	Battery Mgmt System	FR4 PCB
12	STAT-2026-BCK	1	Buck Converter	FR4 PCB
13	STAT-2026-CTL	1	Controls PCB	FR4 PCB